

Lab 3: Component Modelling & Architectural Pattern Selection

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Section: 5F

Course: Software Engineering

Problem Statement Number: 11 (Telemedicine Slot Booking & Prescription Portal)

Written Justification

Architecture Selection: We chose Layered Architecture for the Telemedicine Slot Booking & Prescription Portal system.

Reason 1: The system's functionality naturally separates into three concerns: the patient/physician-facing interface (browsing doctors, booking slots, joining consultations, viewing prescriptions), the core business logic (checking slot availability, generating encrypted room links, authoring and signing prescriptions), and persistent data storage (bookings, prescription records, doctor schedules). This maps directly onto the Presentation, Business, and Data layers of a layered architecture.

Reason 2: The system is a single cohesive application rather than several independently scaled teams or services, so the extra operational complexity of microservices (service discovery, inter-service network calls, distributed data consistency) is not justified here. A layered architecture is simpler to implement, test, and maintain, while still keeping concerns cleanly separated.

Security Advantage: The layered architecture provides clear boundaries for applying security controls. The Data layer can consistently enforce encryption of stored clinical and booking data using AES-256, while communication between layers can use TLS 1.3. Separating these responsibilities also reduces the risk of inconsistent security handling across the system.

Performance Benefit: The layered architecture can reduce communication overhead when components within the business layer are deployed together. For example, booking and scheduling operations can interact with the video consultation logic without requiring the distributed service-to-service network calls that would typically be present in a microservices architecture.